

Vegetable and fruit intake after diagnosis and risk of prostate cancer progression

Cruciferous vegetables, tomato sauce, and legumes have been associated with reduced risk of incident advanced prostate cancer. In vitro and animal studies suggest these foods may inhibit progression of prostate cancer, but there are limited data in men. Therefore, we prospectively examined whether intake of total vegetables, and specifically cruciferous vegetables, tomato sauce, and legumes, after diagnosis reduce risk of prostate cancer progression among 1,560 men diagnosed with non-metastatic prostate cancer and participating in the Cancer of the Prostate Strategic Urologic Research Endeavor, a United States prostate cancer registry. As a secondary analysis, we also examined other vegetable sub-groups, total fruit, and subgroups of fruits. The participants were diagnosed primarily at community-based clinics and followed from 2004–2009. We assessed vegetable and fruit intake via a semi-quantitative food frequency questionnaire, and ascertained prostate cancer outcomes via urologist report and medical records. We observed 134 events of progression (53 biochemical recurrences, 71 secondary treatments likely due to recurrence, six bone metastases, four prostate cancer deaths) during 3,171 person-yrs. Men in the fourth quartile of post-diagnostic cruciferous vegetable intake had a statistically significant 59% decreased risk of prostate cancer progression compared to men in the lowest quartile (hazard ratio (HR): 0.41; 95% confidence interval (CI): 0.22, 0.76; p-trend: 0.003). No other vegetable or fruit group was statistically significantly associated with risk of prostate cancer progression. In conclusion, cruciferous vegetable intake after diagnosis may reduce risk of prostate cancer progression.